

What is Networking?

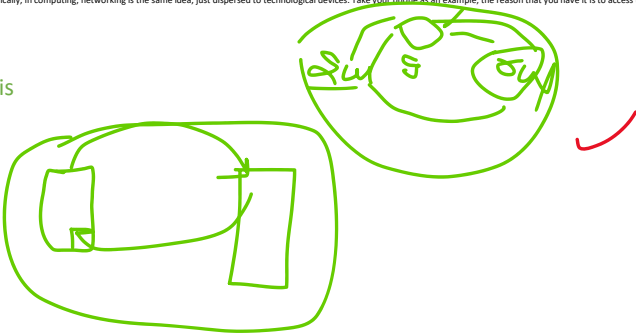
Networks are simply things connected. For example, your friendship circle: you are all connected because of similar interests, hobbies, skills and sorts.

Networks can be found in all walks of life:

- A city's public transportation system
- Infrastructure such as the national power grid for electricity
- Meeting and greeting your neighbours
- Postal systems for sending letters and parcels

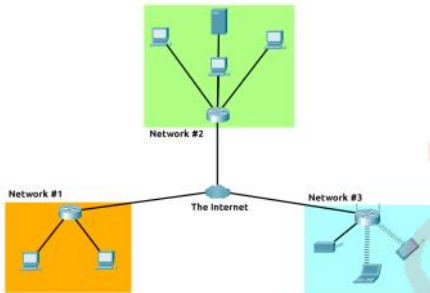
But more specifically, in computing, networking is the same idea, just dispersed to technological devices. Take your phone as an example: the reason that you have it is to access things. We'll cover how these devices communicate with each other and the rules that follow.

What is



the Internet?

Now that we've learnt what a network is and how one is defined in computing (just devices connected), let's explore the Internet. The Internet is one giant network that consists of many, many small networks within itself. Using our example from the previous task,



As previously stated, the Internet is made up of many small networks all joined together. These small networks are called private networks, where networks connecting these small networks are called public networks -- or the Internet! So, to recap, a network can be one of two types:

- A private network
- A public network

Identifying Devices on a Network

Devices on a network are very similar to humans in the fact that we have two ways of being identified:

- Our Name
- Our Fingerprints

Now we can change our name through deed poll, but we can't, however, change our fingerprints. Every human has an individual set of fingerprints which means that even if they change their name, there is still an identity behind it. Devices have the same thing: two means of identification, with one being permeable. These are:

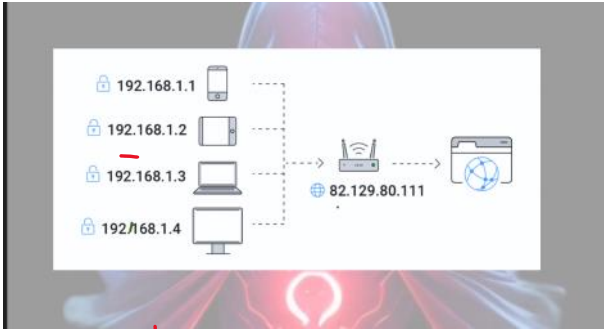
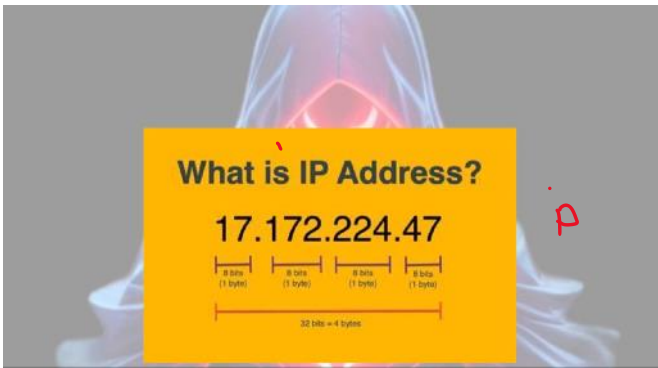
- An IP Address
- A Media Access Control (MAC) Address -- think of this as being similar to a serial number.



IP Addresses

Briefly, an IP address (or Internet Protocol) address can be used as a way of identifying a host on a network for a period of time, where that IP address can then be associated with another device without the IP address changing. First, let's split up precisely what an IP address is in the diagram below:



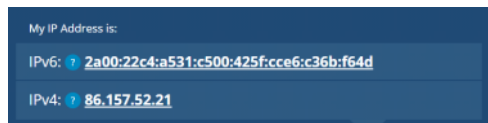


Cisco, an industry giant in the world of networking, estimated that there would be approximately 50 billion devices connected on the Internet by the end of 2021. [Cisco, 2021](#). Enter IP address versions. So far, we have only discussed one version of the Internet Protocol addressing scheme known as IPv4, which uses a numbering system of 2^{32} IP addresses (4.29 billion) -- so you can see why there is such a shortage!

IPv6 is a new iteration of the Internet Protocol addressing scheme to help tackle this issue. Although it is seemingly more daunting, it boasts a few benefits:

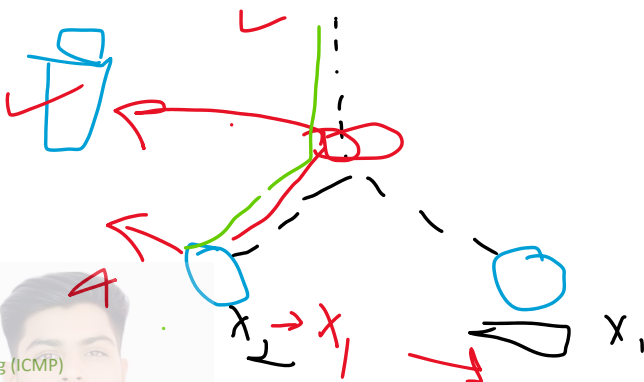
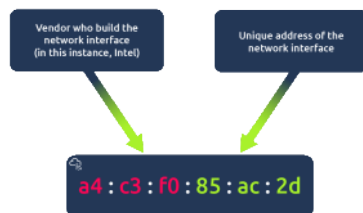
- Supports up to 2^{128} of IP addresses (340 trillion-plus), resolving the issues faced with IPv4
- More efficient due to new methodologies

The screenshot below compares both an IPv6 and IPv4 address.



MAC Addresses

Devices on a network will all have a physical network interface, which is a microchip board found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC (Media Access Control)** address. `04:c3:f0:85:ac:2d`. The first six characters represent the company that made the network interface, and the last six is a unique number.

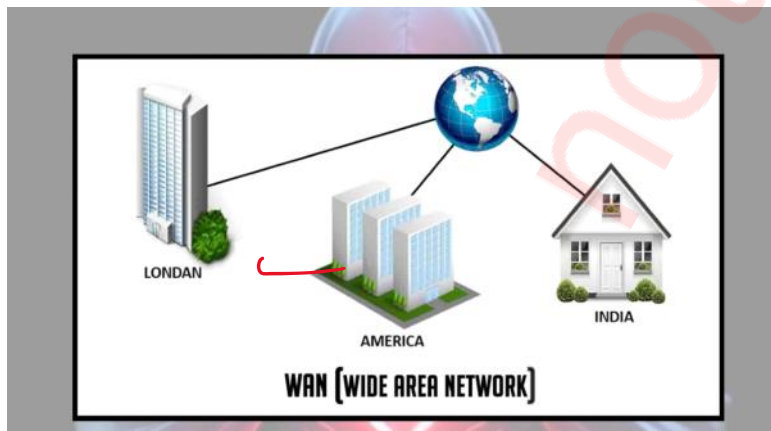
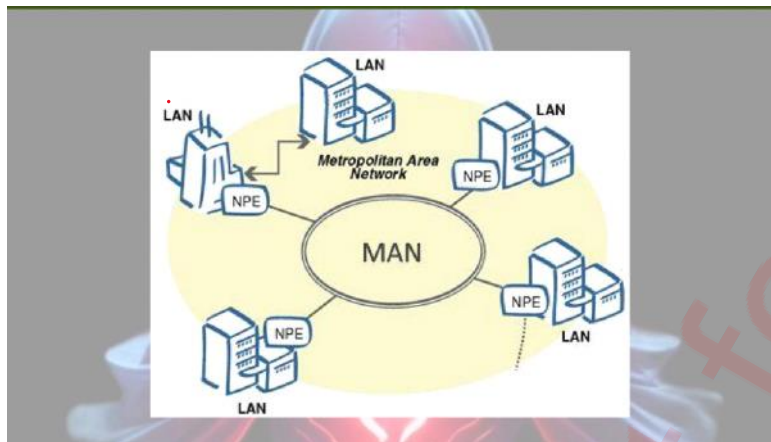
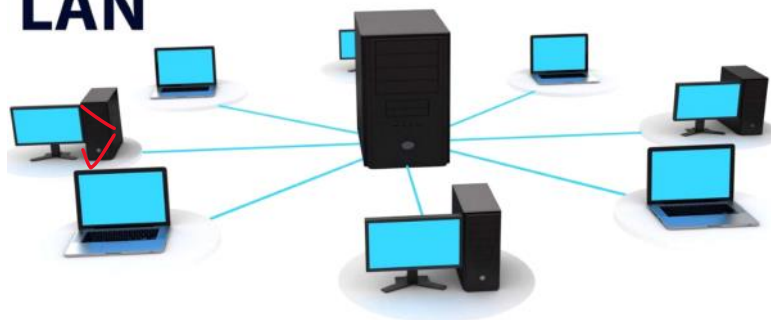


Ping (ICMP)

Ping is one of the most fundamental network tools available to us. Ping uses ICMP (Internet Control Message Protocol) packets to determine the performance of a connection between devices, for example, if the connection exists or is reliable. The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Type of networks

LAN



Introducing LAN Topologies

Local Area Network (LAN) Topologies

Over the years, there has been experimentation and implementation of various network designs. In reference to networking, when we refer to the term "topology", we are actually referring to the design or look of the network at hand. Let's discuss the advantages and disadvantages of these topologies below.

Star Topology

The main premise of a star topology is that devices are individually connected via a central networking device such as a switch or hub. This topology is the most commonly found today because of its reliability and scalability—despite the cost.

Bus Topology

This type of connection relies upon a single connection which is known as a backbone cable. This type of topology is similar to the leaf off of a tree in the sense that devices (leaves) stem from where the branches are on this cable.

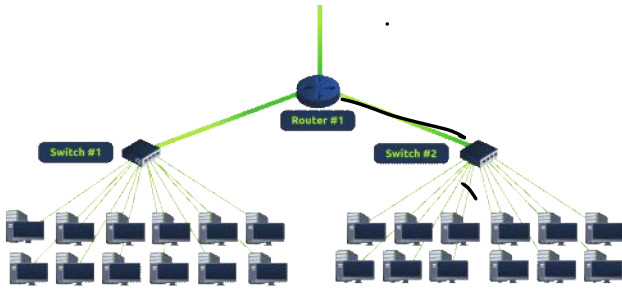
Ring Topology

The ring topology (also known as token topology) boasts some similarities. Devices such as computers are connected directly to each other to form a loop, meaning that there is little cabling required and less dependence on dedicated hardware such as within a star topology.

What is a Switch?

Switches are dedicated devices within a network that are designed to aggregate multiple other devices such as computers, printers, or any other networking-capable device using ethernet.

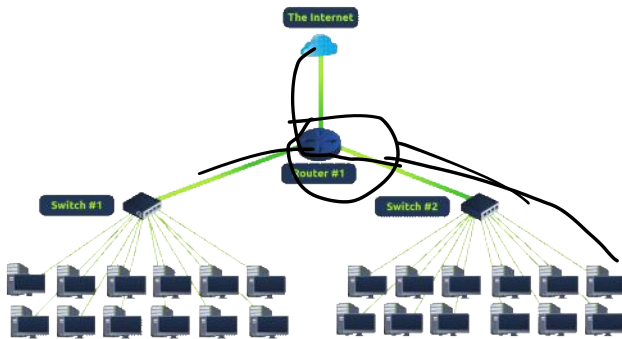




Both Switches and Routers can be connected to one another. The ability to do this increases the redundancy (the reliability) of a network by adding multiple paths for data to take. If one path goes down, another can be used. Whilst this may reduce the overall performance of a network because packets have to take longer to travel, there is no downtime – a small price to pay considering the alternative.

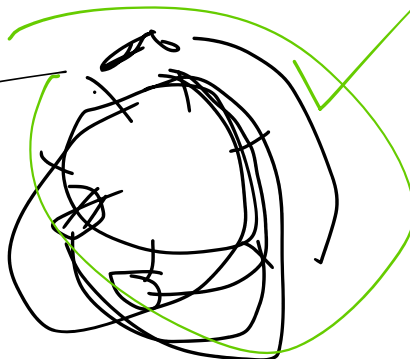
What is a Router?

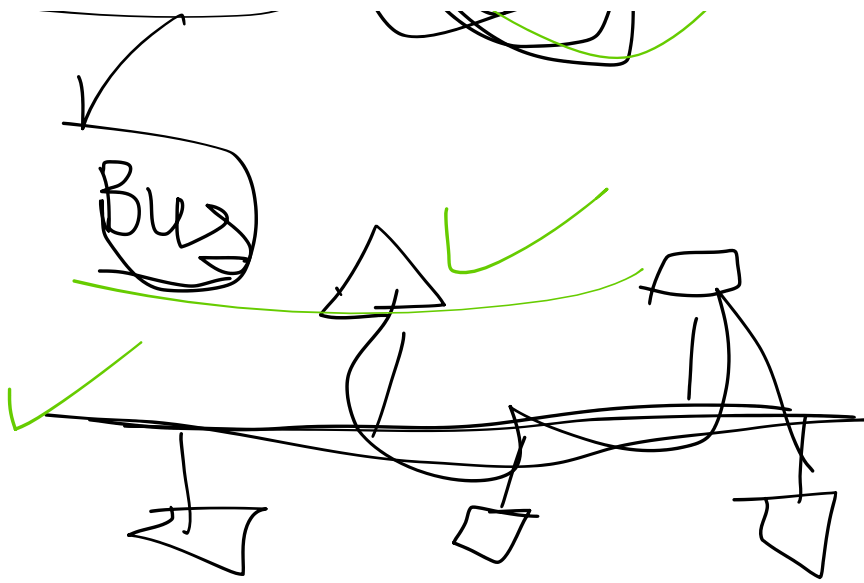
It's a router's job to connect networks and pass data between them. It does this by using routing (hence the name router!). Routing is the label given to the process of data travelling across networks. Routing involves creating a path between networks so that this data can be successfully delivered.



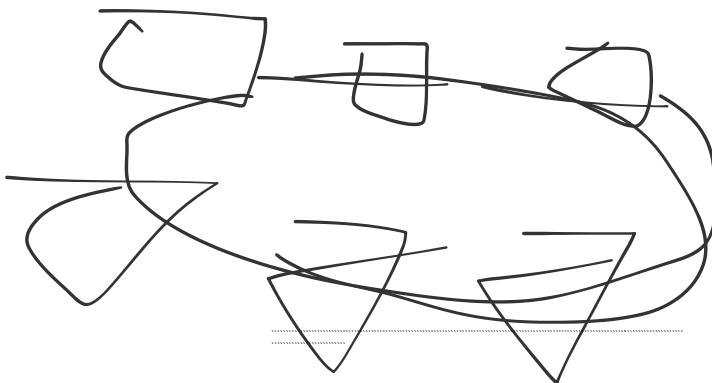
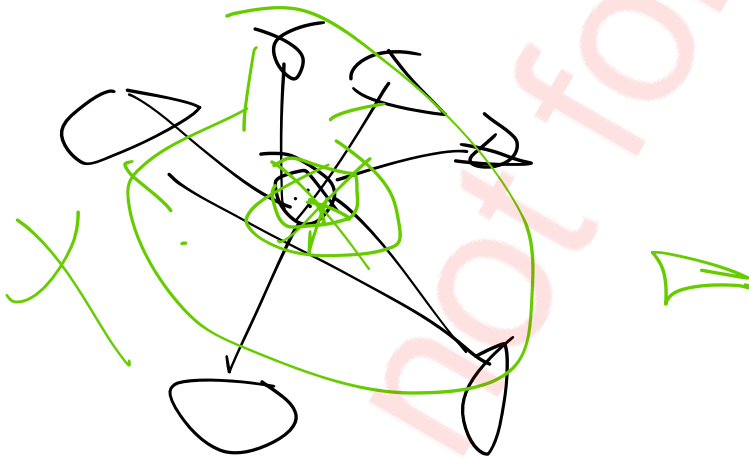
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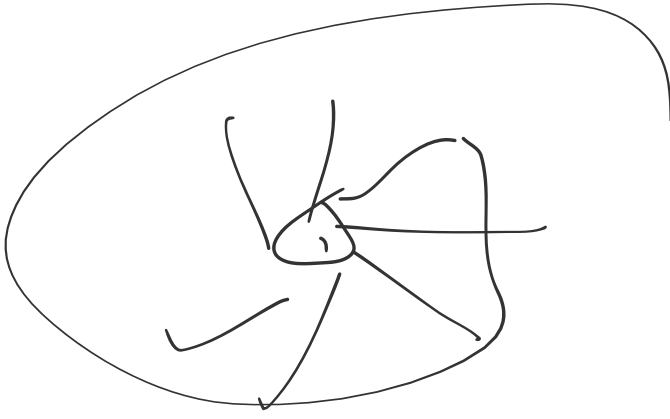
Ring ✓





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